
Product Requirement Document

Clacker Tracker

Minimalistic Electronic Click Logger – on-Device CSV – Mobile Companion Software

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PRD + Build Requirements	1.0 (MVP)	Draft for Review	August 2026

Purpose: Define everything required to design, build, code, and ship Clacker Tracker: a minimalistic hardware click logger with on-device CSV storage and mobile software for data management and multi-project organization

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Executive Summary & Product Vision

Clacker Tracker is a small, battery-powered electronic device whose sole job is to detect physical clicks from the user and record each click with an accurate timestamp into an on-device CSV log. It maintains a running daily total and a continuous background clock (date + time). A companion mobile application (phone/tablet) allows the user to import or export the CSV data, assign clicks or sessions to named projects, and manage multiple projects.

The product is intentionally minimal: no cloud requirement for core use, no continuous BLE streaming, no complex sensors. The hardware owns the reliable log; the software owns organization, analysis, and portability of that log. This PRD covers the complete build needs - hardware, firmware/code, and mobile software - so a small team or individual can take the product from concept to working prototype.

Problems, Goals & Success Criteria

Manual counting (mental tally, paper, basic mechanical clickers) loses history, lacks precise timestamps, and cannot be organized by project. Phone-based counters interrupt flow and drain battery. Clacker Tracker solves this by providing a dedicated physical click surface that silently logs every event with real time, while the mobile software turns the raw log into project-aware, exportable data.

2.1 Product Goals

- Capture every user click with millisecond-accurate (or better) timestamp and persist it in an on-device CSV file.
- Maintain independent daily click totals and a continuous background date/time clock.
- Enable the user to transfer the log to a phone and assign data to multiple named projects.
- Keep the hardware simple, low-power, and durable for everyday carry.

2.2 Success Metrics (MVP)

Metric	Target	How Measured
Missed / double-counted clicks	< 0.5 % under normal use	Lab + field button test
Timestamp accuracy	±2 seconds after 7 days (or better with RTC)	Compare device clock vs reference
Battery life (typical use)	≥ 6 months on primary cell	1-50 clicks/day, occasional export
CSV integrity	Zero corrupted rows after 10 k clicks	File system + parse validation
Export success (first attempt)	≥ 95 %	User testing of transfer flow

Personas & Uses

Clacker Tracker is useful for anyone who needs a reliable physical way to log discrete clicks and later organize those clicks by project or purpose. It is especially valuable when the counting activity itself should not require looking at or unlocking a phone.

3.1 Primary Personas & Use Cases

Persona	How they use this device
Project / Task Tracker	Clicks completed tasks or work units; later assigns each day's (or session's) clicks to a named project in the mobile app for time/effort logging.
Habit / Productivity User	Logs daily habits or pomodoros with a single physical click. Reviews totals and exports CSV for personal analytics.
Maker / Crafter / Tester	Counts stitches, layers, test cycles, or assembly steps. Imports log and tags by project or experiment name.
Fitness / Rehab	Records reps or therapy movements without interrupting form. Daily totals and project tags (e.g., "left leg", "week 3") live in the app.
Field / Light Industrial	Spot-counts inventory or quality checks; later imports CSV and assigns to job numbers or locations as projects.

3.2 Additional Uses

- Classroom participation or response counting; research event logging; board-game or hobby score keeping; simple accessibility click logging.
- Anti-personas (out of MVP scope): continuous GPS tracking, multi-user enterprise dashboards, high-speed industrial counting (>10 Hz sustained), safety-critical systems.

What We Need – Complete Requirement

The product has three coordinated parts: (A) the physical electronic device, (B) the firmware/code that runs on the device, and (C) the mobile software. Below is everything required for each.

4.1 Hardware — What to Design & Source

- Microcontroller: Nordic nRF52840 (or equivalent with BLE + enough flash/RAM for CSV logging and filesystem). Provides low power, BLE for data transfer, and sufficient storage.
- Click input: high-quality tactile switch (or membrane) rated for $\geq 100\,000$ cycles; mechanical debounce handled in firmware.
- Real-time clock: either MCU internal RTC with crystal or external low-power RTC module so date/time continues accurately while the device sleeps.
- Storage: onboard flash (or external SPI NOR/NAND) large enough for months of CSV rows (each row $\approx 30\text{--}50$ bytes). Prefer a simple filesystem (LittleFS or FatFS).
- Power: CR2032 or small Li-ion + charging; power path designed for multi-month life at low click rates.
- User feedback: single LED (or RGB) for power-on, click confirmation, low-battery, and transfer mode.
- Enclosure: minimalistic, pocketable or clip-on, IP54 recommended; one button + LED visible.
- Connectivity for data transfer: BLE (preferred for wireless CSV pull) or USB-C mass-storage / serial. BLE is the primary path for the mobile software.

4.2 Firmware / Device Code — What to Write

- Boot & initialization: configure clocks, RTC, GPIO for button and LED, filesystem mount.
- Background clock: maintain continuous date and time (year-month-day, hour:minute:second) that survives sleep and power cycles (battery-backed RTC).
- Click detection: interrupt or polled reading of the button with software debounce (≤ 20 ms); on valid click, capture current timestamp and increment daily counter.
- CSV logging: append one row per click to a file (example columns: timestamp_iso, unix_epoch, daily_total_after_click, optional session_id). Create new file or rotate daily if desired.
- Daily totals: keep a running count for the current calendar day; reset at midnight according to the RTC.
- Low-power modes: deep sleep between clicks; wake on button interrupt; wake for periodic clock maintenance if needed.
- Data transfer interface: expose the CSV file(s) over BLE (custom GATT or Nordic UART / file transfer profile) or as USB mass storage so the mobile app can import the log.
- Safety: wear-leveling or safe append so power loss mid-write does not corrupt the entire log; simple integrity check on boot.

- Optional: simple on-device menu or long-press actions (e.g., force new session, clear daily total) if hardware allows.

4.3 Mobile Software — What to Build

- Platform: native or cross-platform (Flutter / React Native / Kotlin Multiplatform) for iOS and Android.
- Import: discover the Clacker Tracker via BLE (or USB), request the CSV log, parse rows, and store them in local app database.
- Export: allow user to export selected data (or whole project) as CSV / JSON / share sheet for external tools.
- Projects: create, rename, archive multiple named projects; assign imported click rows or whole days/sessions to one or more projects.
- Views: daily totals, project summaries, simple charts (clicks over time), raw log browser with search/filter by date or project.
- Time sync (optional but recommended): when connected, allow the app to set or correct the device RTC so timestamps stay accurate long-term.
- Local-first: all core features work offline; no mandatory account or cloud for MVP.
- Permissions & UX: clear Bluetooth/USB permission flows, progress indicators during transfer, confirmation of successful import.

On-Device Data Model & CSV Format

The device is the source of truth for the raw click log. A recommended minimal CSV schema (one row per click):

Column	Example	Description
timestamp	2026-08-07T14:32:11	ISO-8601 local or UTC timestamp of the click
epoch	1754564531	Unix epoch seconds (optional but useful)
daily_total	47	Running total of clicks for that calendar day after this click
seq	1842	Optional monotonic sequence number for integrity

Daily total is also kept in a small separate status record or file so the device can show or report the current day's count without parsing the whole CSV. The mobile app is responsible for mapping rows to projects after import; the device itself does not need to know project names

Core User Flows

6.1 Everyday Click Logging (no phone needed)

- User presses the button → device wakes, debounces, reads RTC, appends CSV row, increments daily total, flashes LED, returns to sleep.
- At midnight (RTC) the daily total resets; a new day's logging continues seamlessly.

6.2 Import & Project Assignment

- User opens mobile app → connects to Clacker Tracker → initiates import → app pulls CSV → parses and stores rows → user selects a project (or creates one) and assigns the imported data (whole file, date range, or individual rows).
- User can later export any project's data as CSV for spreadsheets, research, or archiving.

6.3 Time Maintenance

- On each successful connection the app may offer to sync the device clock to phone time, correcting any drift.

Non-Functional Requirements & Constraints

Area	Requirement
Power	Average current low enough for ≥ 6 months on a CR2032 (or equivalent) at typical click rates; deep sleep between events.
Reliability	No lost clicks across 10 000 presses; CSV remains parseable after unexpected power loss.
Storage	Enough flash for at least 3–6 months of continuous daily use before the user must import/clear.
Privacy	All click data stays on device and on the user's phone; no mandatory cloud or telemetry in MVP.
Cost target	Electronics BOM aiming for low double-digit USD at modest volume so the finished product can stay affordable.
Compliance	FCC/CE (or local equivalent) for intentional radiator if BLE is used; RoHS; battery safety labels.

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